

A Dynamic Programming Characterization of Optimal Bidding in Stochastic Sequential Auctions

Varnit Sahu

Problem

Bidders face a sequence of auctions over time and must allocate a finite budget strategically across stages.

In this article, we study a simplified stochastic multi-stage auction model and derive a recursive characterization of the optimal bidding strategy using dynamic programming. The goal is to show that, under minimal assumptions, optimal behavior can be described by a value function depending only on the remaining budget and the current stage.

Model Setup

We consider a finite horizon of T stages, indexed by $t = 1, 2, \dots, T$. At each stage, a single whole item is auctioned using a first-price auction.

The bidder we study has:

- A remaining budget $s \geq 0$,
- A private valuation v_t for the item at stage t , where v_t is a non-negative random variable with known distribution,
- A bidding decision $b_t \in [0, s]$.

Let M_t denote the highest competing bid from other bidders at stage t . We assume M_t is a non-negative random variable with known cumulative distribution function $F_t(b) = \mathbb{P}(M_t \leq b)$.

The bidder wins the item at stage t if $b_t \geq M_t$, pays b_t , and receives utility v_t .

Expected Utility Decomposition

Let us fix a stage t and remaining budget as s . Suppose the bidder observes valuation $v_t = v$ and submits a bid $b \in [0, s]$.

There are two possible outcomes:

- With probability $F_t(b)$, the bidder wins the auction, receives value v , pays b , and continues to stage $t + 1$ with budget $s - b$.
- With probability $1 - F_t(b)$, the bidder loses the auction and continues with budget s .

Let $V_t(s)$ denote the maximum expected utility achievable from stages t through T given remaining budget s .

Then the expected utility of bidding b at stage t is:

$$Q_t(s, v, b) = F_t(b)(v + V_{t+1}(s - b)) + (1 - F_t(b))V_{t+1}(s).$$

Lemma: Linearity of Expectation

Lemma. For fixed t and s , the expected utility $Q_t(s, v, b)$ is affine in the win probability $F_t(b)$.

Proof. The expression for $Q_t(s, v, b)$ is a convex combination of the win and loss utilities, weighted by $F_t(b)$ and $1 - F_t(b)$. This follows directly from the linearity of expectation. $\square$

Bellman Optimality Equation

Since future decisions depend only on the remaining budget and the current stage, the optimal value function satisfies the following recursion.

Proposition (Bellman Equation). For $t = 1, 2, \dots, T$, the value function satisfies:

$$V_t(s) = \mathbb{E}_{v_t} \left[\max_{0 \leq b \leq s} (F_t(b)(v_t + V_{t+1}(s - b)) + (1 - F_t(b))V_{t+1}(s)) \right],$$

with terminal condition $V_{T+1}(s) = 0$ for all $s \geq 0$.

Proof. At stage t , the bidder observes valuation v_t and chooses a bid $b \in [0, s]$ to maximize expected utility. By conditioning on the auction outcome and applying the lemma above, the expected utility of any bid is given by $Q_t(s, v_t, b)$. Optimizing over b yields the optimal continuation value for a fixed v_t . Taking expectation over the valuation distribution completes the recursion. $\square$

Interpretation

The Bellman equation shows that optimal bidding behavior is *Markovian*: the bidder does not need to remember past bids or outcomes, only the remaining budget and current stage.

This recursive structure enables efficient computation of optimal bidding policies through backward induction, after discretizing the budget space if necessary.

Conclusion

We have derived a dynamic programming characterization of optimal bidding in a finite-horizon stochastic auction with budget constraints. Despite the strategic nature of the problem, the solution relies only on elementary probabilistic reasoning and recursive optimization.